

Jivan Singh

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EDUCATION

Illinois Institute of technology | Chicago, IL, United States
Master of Science in Computer Science

Expected May 2026
GPA: 3.40/4.00

University of Mumbai | Mumbai, India
Bachelor of Engineering in Computer Engineering

May 2023
GPA: 3.80/4.00

PROFESSIONAL EXPERIENCE

Software Developer | Crayon Infotech Pvt Ltd., India

Aug 2023 – Jul 2024

Tech Stack: PHP | CodeIgniter | Laravel | Python | Git | SQL/NOSQL | PostgreSQL | RESTful API | Flask API | Data Structures

- Designed scalable software solutions, leveraging techniques to enhance performance by 30% and cut downtime by 15%
- Developed robust APIs for seamless third-party integration, boosting functionality and enhancing the user experience
- Collaborated with teams to deploy 5+ applications, refining the codebase to ensure stability and minimize bugs
- Engineered and maintained backend systems using Python, Flask, and PostgreSQL, optimizing data flow, improving query performance, and ensuring high availability and scalability for real-time web applications

Blockchain Research Intern | Lab Systems Pvt. Ltd., India

Jul 2022 – Feb 2023

Tech Stack: Blockchain Architecture | Blockchain Node | Hashing | Solidity | Smart Contracts | Ganache | IPFS

- Gained expertise in cryptocurrency forensics using OSINT skills, training a team to develop a comprehensive solution
- Managed a Bitcoin Node to support research initiatives, actively contributing to blockchain technology advancements
- Trained a team of 12 professionals in blockchain technology, enhancing leadership capabilities and technical expertise

Python Intern | Jeet Techno Solutions LLP, India

May 2019 – Jun 2019

Tech Stack: Python | Tkinter | OOP | File Handling | GUI Programming | Data Structures

- Built and deployed Python desktop applications using GUI, applying design principles and modular programming
- Assisted in building embedded software for real-time data monitoring, integrating sensor feedback and user interaction
- Collaborated with developers and trainers in the software development lifecycle from planning to client demonstration
- Conducted stakeholder interviews and on-site research to gather functional and technical requirements, contributing to user-centric feature development and successful project validation

PROJECTS

Time-Series Forecasting | *Tech Stack: TensorFlow/Keras, Facebook Prophet, StatsModels, Flask, React, Docker, GCP*

- Built a Python-Flask pipeline to ingest GitHub data (issues, PRs, commits, branches, contributors, releases) from 2 years
- Engineered robust microservices for model training and forecasting with a React dashboard, containerized using Docker
- Deployed on Google Cloud Run with automated CI/CD, improving system efficiency, scalability, and reliability by 40%

Graphical Password Representation using Blockchain | *Tech Stack: Ethereum Blockchain, IPFS, Hardhat, React.js*

- Constructed a blockchain solution that replaces traditional passwords by using smart contracts and IPFS integration
- The solution enhances user authentication by decentralizing data storage, providing greater security and improved privacy

Pulse Detection through webcam | *Tech Stack: Python, OpenCV, Tkinter, NumPy, SciPy*

- Implemented a Python application with a Tkinter interface, utilizing Laplace Pyramid and K-Nearest Neighbour
- Achieved high accuracy in predictions by implementing algorithms like Fast Fourier Transform, attaining an R^2 of 0.9

SKILLS & CERTIFICATIONS

- Programming Languages:** Python, C++/C, Java, JavaScript, TypeScript, SQL, NoSQL, PHP, HTML, CSS, SASS
- Frameworks & Libraries:** Next.js, React.js, Node.js, Express.js, CodeIgniter, Laravel, Flask, Django, Tkinter, Pandas, NumPy, Matplotlib
- Tools & Platforms:** Git/GitHub, Docker, CI/CD (GitHub Actions), AWS (EC2, S3, Lambda), Firebase, Postman, REST APIs, JIRA, Slack, Vercel
- Systems & Concepts:** Data Structures & Algorithms, Object-Oriented Programming, System Design, Operating Systems, Networking, Agile Methodologies, Software Project Management, API Integration
- Certifications:** AI Agents in LangGraph (DeepLearning.AI), AI Agentic Design Patterns with AutoGen (DeepLearning.AI), AWS Academy Cloud Foundations, Advanced Software Engineering (Walmart USA)

TECHNICAL ACHIEVEMENTS

- Runner-up in Internal Smart India Hackathon 2020

PATENT & PUBLICATIONS

- Medilog** – (Granted), German Patent File No - 20 2023 102 823.3 | Granted: September 19, 2023
Delivered a blockchain-based web app for tracking prescription drugs, improving supply chain transparency and traceability by 80%, while reducing counterfeit incidents in pilot use. [Patent Link](#)
- Published “Construction Estimation Project” in **IJSRD**. [View Paper](#)
- Published “Pulse Detection Through Webcam” in **IJRAR**. [View Paper](#)